

Geochem Water Sampling Protocol

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Sampling Parameters

Purge Volumes: (Low end of range is for shallow wells (~60 cm) during wet season and all wells in dry season; high end of range is for deep wells in wet season)

- 250-500 mL for 1" (diameter) piezometers
- 1000-2000 mL for 2" piezometers

Sampling Rate: Purging can be done at slow-moderate pump rate; sampling at slow pump rate.

In-Field Measurements

pH, temp, ORP, EC: Use multi-parameter probe. Calibrate pH w/pH 4, 7, 10 buffers and Conductivity (EC) w/1413 $\mu\text{S}/\text{cm}$ solution. Temperature is only accurate if measured on a flowing sample or directly in water.

Chemets: DO, Fe(II), sulfide (optional). Use chemet meter and follow instructions in kit. All samples **except DO** (especially for low range DO) should be filtered.

Water Level: (optional) Record depth to water table from top of well casing (TOC) for groundwater. Measure TOC to ground surface once per season or if casing is modified.

Bottles to Collect

CATION (ICPMS) Sample: 20-mL plastic scintillation vial

Filter (0.2 or 0.45 μm PVDF or Supor filter) water into 20-mL plastic scintillation vial. Fill to shoulder of vial. Sample can be shipped to lab and acidified at LBL with 200 μL ultrex grade nitric acid (8M or concentrated). Samples can be refrigerated, but don't need to be.

ANION (IC) and NH₄ Sample: 20-mL plastic scintillation vial

Filter (0.2 or 0.45 μm PVDF or Supor filter) water into 20-mL plastic scintillation vial. Fill to about $\frac{3}{4}$ full. Freeze sample as soon as possible (end of day) and ship on dry ice. Freezing is necessary to preserve nitrogen species; if not frozen, other anions (Cl, SO₄) can still be measured, but N-species will be inaccurate. Samples may need to be re-filtered prior to analysis if Fe precipitates have formed (for Fe-reducing condition samples).

Acidify sample with 100 μL of 2 M HCl (per 10 mL sample) prior to NH₄ analysis. IC samples should NOT be acidified.

TICTOC Sample: 40-mL amber glass vial with silicone/PTFE septa lid

Filter (0.2 or 0.45 μm PVDF or Supor filter) water into 40-mL amber glass vial with silicone/PTFE septa lid. Fill completely, leaving no headspace. Refrigerate sample as soon as possible and ship with regular ice or ice packs. Do not freeze.

Chemetrics Kits Method #s

Chemetrics method numbers for commonly used test kits. **Note that these method numbers are only valid for the test kit noted in the table.** Consult the manual for other method #s.

Analyte	Kit #	Method #	Analysis Time
Iron	K-6203	100	
Dissolved oxygen, high range	K-7513	141	2 min
Dissolved oxygen, low range	K-7553	142	1 min
Sulfide	K-9503	179	5 min

Part Numbers for ordering more supplies – Available through EBuy

TICTOC vials: VWR# 89093-870, 40-mL amber glass vials – EPA vials with PTFE/silicone septa

CATION/ANION vials: VWR# 66021-690, 20-mL HDPE vials with polyethylene flat caps

Filters: [standard filter] EMD Millipore# SLHV033NK, 0.45 μm Millex-HV PVDF filters, 33mm, non-sterile, pack of 1000 [comes in various types of packs (50-1000) and sterile or non-sterile, just make sure filter membrane type (PVDF), size (0.45 μm) and diameter (33 mm) are the same].

or

[non-standard – make note if 0.22 μm filters used] EMD Millipore# SLGV033RB, 0.22 μm Millex-GV PVDF filters, 33mm, gamma-sterilized, pack of 250 [comes in various types of packs and sterile or non-sterile, just make sure filter membrane type (PVDF), size (0.22 μm), and diameter (33 mm) are the same].

Chemet Kits: (see kit #s above)